

Intermediate Algebras over Nonseparable Compacts Fail BCP

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Definitions

A normed space X has the ball-covering property, or BCP, if its unit sphere is covered by countably many open balls none of which contains the origin. Equivalently, X has BCP if and only if there are countably many centres $x_n \in X$ such that

$$S_X \subseteq \bigcup_n B(x_n, \|x_n\|).$$

Indeed, any open ball not containing the origin has radius strictly smaller than the norm of its centre, and enlarging it to radius $\|x_n\|$ preserves the open-ball escape test.

Statement

Theorem 1 (intermediate nonseparable compact obstruction). *Let H be a nonseparable Hilbert space. Let A be a C^* -algebra such that*

$$\mathcal{K}(H) \subseteq A \subseteq \mathcal{B}(H).$$

Then A fails BCP.

Corollary 1 (faithful nonseparable Type I representation obstruction). *Let A be a C^* -algebra with a faithful irreducible representation $\pi : A \rightarrow \mathcal{B}(H)$ on a nonseparable Hilbert space. If $\mathcal{K}(H) \subseteq \pi(A)$, then A fails BCP.*

Idea of the Proof

The finite-dimensional packet proved failure for algebras of the form $\mathcal{K}(H) + F$ by using the fact that compact perturbations have separable range. For an arbitrary intermediate algebra, the centres can contain general bounded multiplier directions, so the compact-range argument no longer applies directly.

The replacement is to make a separable reducing subspace for the given countable list of centres. For each centre a_n , choose countably many unit vectors that almost norm a_n . Close their span under all words in the operators a_n and a_n^* . The resulting subspace L is separable, reduces every a_n , and still carries the full norm of every a_n .

Since H is nonseparable, $L^\perp \neq 0$. A rank-one projection p inside $L^\perp$ belongs to $\mathcal{K}(H) \subseteq A$. In the decomposition

$$H = L \oplus L^\perp,$$

the operator $p - a_n$ contains the untouched block $-a_n|_L$, whose norm is exactly $\|a_n\|$. Thus p escapes all balls $B(a_n, \|a_n\|)$.

Proof

Lemma 1 (separable norming reducing subspace). *Let H be a Hilbert space, and let $(T_n)_{n \geq 1} \subseteq \mathcal{B}(H)$ be a countable family. There is a separable closed subspace $L \subseteq H$ such that L reduces every T_n and*

$$\|T_n|_L\| = \|T_n\| \quad (n \geq 1).$$

Proof. For each pair $n, k \geq 1$, choose a unit vector $x_{n,k} \in H$ satisfying

$$\|T_n x_{n,k}\| > \|T_n\| - 1/k$$

when $T_n \neq 0$; if $T_n = 0$, choose any unit vector.

Let $\mathcal{W}$ be the countable set of all finite words in the alphabet

$$\{T_n, T_n^* : n \geq 1\},$$

including the empty word. Define

$$L = \overline{\text{span}}\{W x_{n,k} : W \in \mathcal{W}, n, k \geq 1\}.$$

This is separable, because it is generated by countably many vectors. Moreover, L is invariant under every T_j and every T_j^* , since left multiplication by T_j or T_j^* just adds one letter to a word. Thus L reduces every T_j .

Finally, each $x_{n,k}$ belongs to L , so

$$\|T_n|_L\| \geq \sup_k \|T_n x_{n,k}\| = \|T_n\|.$$

The reverse inequality is automatic. Hence $\|T_n|_L\| = \|T_n\|$. $\square$

Proof of Theorem 1. Let (a_n) be an arbitrary countable family of centres in A . Apply Lemma 1 to the same operators regarded as elements of $\mathcal{B}(H)$. We obtain a separable closed subspace $L \subseteq H$ that reduces every a_n and satisfies

$$\|a_n|_L\| = \|a_n\| \quad (n \geq 1).$$

Since H is nonseparable and L is separable, $L^\perp \neq \{0\}$. Choose a unit vector $\xi \in L^\perp$, and let p be the rank-one projection onto $\mathbb{C}\xi$. Then

$$p \in \mathcal{K}(H) \subseteq A, \quad \|p\| = 1.$$

Because L reduces every a_n , relative to

$$H = L \oplus L^\perp$$

we have

$$a_n = a_n|_L \oplus a_n|_{L^\perp}, \quad p = 0 \oplus p.$$

Therefore

$$p - a_n = (-a_n|_L) \oplus (p - a_n|_{L^\perp}),$$

and hence

$$\|p - a_n\| = \max\{\|a_n|_L\|, \|p - a_n|_{L^\perp}\|\} \geq \|a_n|_L\| = \|a_n\|.$$

Thus $p \in S_A$ is outside every open ball $B(a_n, \|a_n\|)$.

Since the countable family (a_n) was arbitrary, no countable collection of BCP balls can cover the unit sphere of A . Hence A fails BCP. $\square$

Proof of Corollary 1. Identify A isometrically with its faithful image $\pi(A)$. Since $\pi(A)$ is an intermediate C^* -algebra between $\mathcal{K}(H)$ and $\mathcal{B}(H)$, Theorem 1 applies. $\square$

Scope and Limitations

This theorem strengthens the earlier packet on $\mathcal{K}(H) + F$, where F was finite dimensional. It shows that no intermediate multiplier algebra between $\mathcal{K}(H)$ and $\mathcal{B}(H)$ can repair BCP when H is nonseparable.

The theorem does not by itself complete the Type I classification. If a general Type I algebra has a nonseparable irreducible representation, that representation may occur only as a quotient representation. Since BCP is not stable under quotients in general, failure of BCP for the represented quotient does not automatically imply failure for the original algebra. The remaining visibility problem is to identify when such quotient-level nonseparable fibres can be lifted into an ambient diagonal escape.

Novelty and Search Bounds

The local run indexes were searched for arXiv:2311.14257, nonseparable compact operators, finite-dimensional extensions, intermediate algebras $\mathcal{K}(H) \subseteq A \subseteq \mathcal{B}(H)$, Type I representations, BCP, and UBCP. Existing packets contained the finite-dimensional extension theorem, the Type I separable-irrep theorem, and the non-Type-I visibility attempt, but no arbitrary intermediate-algebra obstruction.

Web searches on June 17, 2026 for “intermediate C^* -algebra $K(H) \subseteq B(H)$ ball-covering property”, “nonseparable compact operators ball-covering property C^* -algebra”, and “ball-covering property $B(H)$ nonseparable” located nearby source and Calkin/operator-space BCP material, but no exact statement of Theorem 1. The theorem is recorded as a new agent-produced partial obstruction, subject to human literature review.

Human Review Focus

Reviewers should check:

1. the construction of the separable reducing subspace in Lemma 1;
2. the equality $\|a_n|_L\| = \|a_n\|$;
3. the block-diagonal norm computation for $p - a_n$;
4. the use of the equivalent BCP formulation with balls $B(a_n, \|a_n\|)$.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Run `fa_banach_001`, lane 19, *Finite-Dimensional Extensions of Nonseparable Compact-Operator Algebras Fail BCP*, solution packet `2311.14257_nonseparable_compact_operators_fail_bcp`, 2026.
- [3] Run `fa_banach_001`, lane 19, *Type I C^* -Algebras and the Ball-Covering Property*, solution packet `2311.14257_type_i_separable_irrep_bcp`, 2026.